

Discrete-Time Domain Modeling and Design for AC Machine Current Regulation

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Abstract—Complex vector current regulators provide robust performance at high fundamental frequency operation, but there are difficulties in increasing the current regulation bandwidth and in operating at very high fundamental frequencies. In this paper, a discrete-time domain model of general AC machines is proposed. Based on this model, a discrete-time domain current regulator design methodology is proposed. This method is a generalization of the complex vector current regulator's pole-zero cancellation approach and it furthermore explicitly incorporates and compensates the control delay common in discrete-time implementations.

Keywords— Current Regulator; Discrete-time domain AC machine modeling; Discrete-time domain complex vector current regulator

I. INTRODUCTION

Current regulators of AC machines have been designed in the synchronous reference frame to get zero steady state current tracking error at the fundamental component frequency [1]. But, the cross-coupling of states caused by the reference frame transformation deteriorates the dynamic performance at high fundamental component frequencies. State feedback decoupling of the cross-coupling terms improves dynamic performance, but performance still deteriorates at high fundamental component frequencies. Rather than decoupling the cross-coupling by state feedback, the internal model control in [2] and the complex vector current regulator approach in [3] and [4] decouple the cross-coupling by canceling the asymmetric complex poles via asymmetric complex controller zeros. These approaches provide robust current regulation performance thus enabling higher fundamental frequency operation. These approaches were generalized to asymmetric machines in state space [5] and via transfer-matrix approach [6].

Despite its desirable attributes, much less attention has been paid to discrete-time domain implementation issues for the complex vector current regulator. The PWM and the digital control delay cause the digital implementation to deviate from the continuous-time domain model. The robustness of the system is degraded if a controller designed in the continuous-time domain is simply converted to the discrete-time domain. A compensation method for the time

delay and magnitude attenuation caused by the frame rotation during the control delay and the PWM-VSI were proposed [7].

This paper presents a complete discrete-time domain model of AC machines, including asymmetric machines, using state space methods. The model includes the voltage latch and digital control delay, which is generalized to a fractional step delay. Using this, a systematic design method for the discrete-time domain current regulator is proposed. The proposed technique is a generalization of the asymmetric complex pole-zero cancellation approach of complex vector current regulation which now deals explicitly with the fractional step digital control delay.

II. DISCRETE-TIME DOMAIN MODELING OF AC MACHINES

A conceptual block diagram of an AC machine controlled in the discrete-time domain with sampling and control period T is shown in Fig. 1. Given a synchronous frame voltage vector command $\mathbf{v}_{qd}^e(kT)$ at k -th control step, PWM-VSI provides a stationary frame voltage vector $\mathbf{v}_{qd}^s(t)$ applied to the AC machine during the k -th control step. A synchronous frame continuous-time domain model of an AC machine can be written as a state space representation (1)

$$\mathbf{L} \frac{d}{dt} \mathbf{i}_{qd}^e(t) = -\mathbf{r} \mathbf{i}_{qd}^e(t) - (j\omega(t)) \mathbf{L} \mathbf{i}_{qd}^e(t) + \mathbf{v}_{qd}^e(t) - \mathbf{e}_{qd}^e(t) - \mathbf{d}_{qd}^e(t),$$

$$\text{where } \mathbf{L} := \begin{bmatrix} L_q & L_{qd} \\ L_{dq} & L_d \end{bmatrix}, \mathbf{r} := \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix}, \text{ and } \mathbf{j} := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad (1)$$

with stator resistance r , q-axis inductance L_q , d-axis inductance L_d , and mutual inductances L_{qd} and L_{dq} , where $\mathbf{i}_{qd}^e(t)$, $\mathbf{v}_{qd}^e(t)$, $\mathbf{e}_{qd}^e(t)$, and $\mathbf{d}_{qd}^e(t)$ are synchronous frame vector variables for current, voltage, back-EMF, and disturbance voltage,

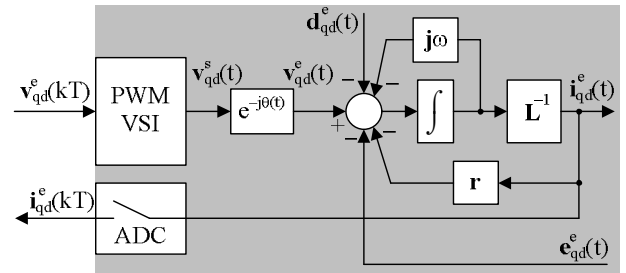


Fig. 1. Conceptual block diagram of a digitally controlled AC machine

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respectively, and where $\omega(t)$ and $\theta(t)$ are the reference frame velocity and position. For fast sample/switching rates, it is reasonable to assume that the frame rotation velocity $\omega(t)$ during a control period T is approximately constant, thus

$$\theta(t) = \theta(kT) + \omega(t - kT). \quad (2)$$

A. The Stationary Frame Voltage Latch Model

The output voltage of a PWM-VSI during a switching period contains modulation harmonics, which depend on the commanded voltage magnitude and angle. However, for discrete-time modeling, it is reasonable to treat the output voltage of a PWM-VSI as held constant at the commanded voltage vector location (3) during the switching

$$\mathbf{v}_{qd}^s(t) = \mathbf{e}^{j\theta(t)} \cdot \mathbf{v}_{qd}^e(kT), \text{ where } \mathbf{e}^{j\theta} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}, \quad (3)$$

period T , as shown in the block diagram in Fig. 2. It is important to note that the voltage is latched in the stationary frame. Therefore, in the synchronous frame, the input voltage from the PWM-VSI is a vector (4) rotating in the reverse direction from the synchronous frame.

$$\mathbf{v}_{qd}^e(t) = \mathbf{e}^{-j\theta(t)} \cdot \mathbf{v}_{qd}^s(t) = \mathbf{e}^{-j\omega(t-kT)} \cdot \mathbf{v}_{qd}^e(kT) \quad (4)$$

B. The Synchronous Frame Back-EMF Latch Model

For a PM or a controlled field machine, the rotor flux doesn't change if seen from the rotor synchronous frame. Also, for an induction machine, the rotor flux is much slower than the current control period, and can also be assumed to be constant if seen from the rotor flux synchronous frame. Thus, it is reasonable to model the back-EMF as latched in the synchronous frame as shown in Fig. 2 and write

$$\mathbf{e}_{qd}^e(t) = \mathbf{e}_{qd}^e(kT). \quad (5)$$

C. The Discrete-Time Domain Model without Delays

Applying the voltage and back-EMF expressions (3) and (4) to the machine equation (1) and dividing the disturbance input $\mathbf{d}_{qd}^e(t)$ in (1) into two parts whereby one part is latched in the synchronous frame as $\mathbf{d}_{qd-e}^e(kT)$ and the other part is latched in the stationary reference frame as $\mathbf{d}_{qd-s}^e(kT)$, (6) is obtained.

$$\begin{aligned} \mathbf{L} \cdot \frac{d}{dt} \mathbf{i}_{qd}^e(t) = & -\mathbf{r} \cdot \mathbf{i}_{qd}^e(t) - (\mathbf{j}\omega) \mathbf{L} \cdot \mathbf{i}_{qd}^e(t) \\ & + \mathbf{e}^{-j\omega(t-kT)} \cdot [\mathbf{v}_{qd}^e(kT) - \mathbf{d}_{qd-s}^e(kT)] - \mathbf{e}_{qd}^e(kT) - \mathbf{d}_{qd-e}^e(kT). \end{aligned} \quad (6)$$

A solution of the machine differential equation (6) for the k -th

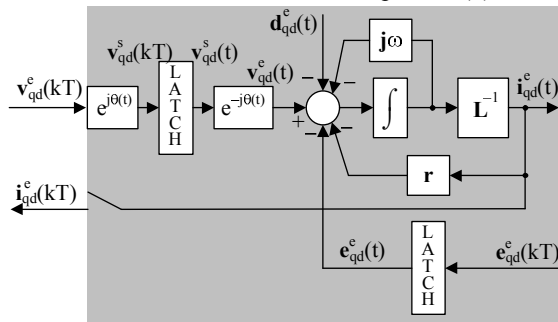


Fig. 2. Block diagram of a digitally controlled AC machine with stationary frame voltage latch and synchronous frame back-EMF latch

sampling period T can be expressed via (7).

$$\begin{aligned} \mathbf{L} \cdot \mathbf{i}_{qd}^e(kT+T) = & -\mathbf{e}^{-(\mathbf{r}+\mathbf{j}\omega\mathbf{L})T} \cdot \mathbf{i}_{qd}^e(kT) \\ & + \int_{\tau=0}^T \mathbf{e}^{-(\mathbf{r}+\mathbf{j}\omega\mathbf{L})(T-\tau)} \left\{ \mathbf{e}^{-\mathbf{j}\omega\tau} \cdot [\mathbf{v}_{qd}^e(kT) - \mathbf{d}_{qd-s}^e(kT)] \right. \\ & \left. - \mathbf{e}_{qd}^e(kT) - \mathbf{d}_{qd-e}^e(kT) \right\} d\tau \end{aligned} \quad (7)$$

By replacing the time domain vector variables with discrete-time domain ones, the following discrete time difference equation model (8) of a general ac machine is achieved.

$$\begin{aligned} \mathbf{i}_{qd}^e[k+1] = & \mathbf{A} \cdot \mathbf{i}_{qd}^e[k] + \mathbf{B} \cdot \mathbf{v}_{qd}^e[k] - \mathbf{B}_e \cdot \mathbf{v}_{qd}^e[k] \\ & - \mathbf{B} \cdot \mathbf{d}_{qd-s}^e[k] - \mathbf{B}_e \cdot \mathbf{d}_{qd-e}^e[k], \end{aligned} \quad (8)$$

where $\mathbf{A} := \mathbf{L}^{-1} \cdot \mathbf{e}^{-(\mathbf{r}\mathbf{L}^{-1}+\mathbf{j}\omega\mathbf{L})T} \cdot \mathbf{L}$,
 $\mathbf{B} := \mathbf{L}^{-1} \cdot \mathbf{e}^{-(\mathbf{r}\mathbf{L}^{-1}+\mathbf{j}\omega\mathbf{L})T} \int_0^T \mathbf{e}^{(\mathbf{r}\mathbf{L}^{-1}+\mathbf{j}\omega\mathbf{L})\tau} \cdot \mathbf{e}^{-\mathbf{j}\omega\tau} \cdot d\tau$
 $\mathbf{B}_e := [\mathbf{r} + (\mathbf{j}\omega)\mathbf{L}]^{-1} [\mathbf{I} - \mathbf{e}^{-(\mathbf{r}\mathbf{L}^{-1}+\mathbf{j}\omega\mathbf{L})T}]$

From the linear system theory, the matrix exponential $\mathbf{e}^{\mathbf{M}}$ is defined as

$$\mathbf{e}^{\mathbf{M}} := \sum_{k=0}^{\infty} \frac{\mathbf{M}^k}{k!}. \quad (9)$$

and the closed form solution of the matrix exponential can be found using the Laplace transform of

$$\mathbf{e}^{\mathbf{M}} = \mathcal{L}\{(s\mathbf{I} - \mathbf{M})^{-1}\} \quad (10)$$

or by using existing numerical solution methods.

D. Discrete-Time Domain Model including Delay

In realistic digital ac machine control, computational delay can not be removed. Thus, digital controls can have up to one-step delay. But, due to improvements in computing performance, the control computation time is getting much shorter than the PWM period. In a symmetric PWM driven system, current sampling in the middle of one PWM cycle gives virtually harmonic-free current measurement. Therefore, a half step control delay can easily be implemented. In this section, an ac machine discrete time model with a fractional step control delay is derived. For this model, a new input voltage vector variable $\mathbf{v}_{qd-m}^e[k]$ is used. It is delayed by $(1-m)$

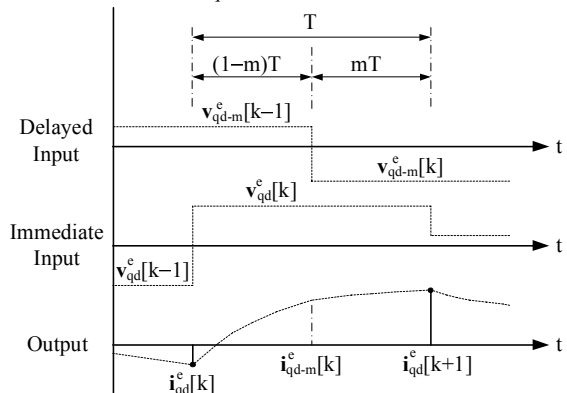


Fig. 3. Conceptual control timing diagram with both delayed input and immediate input with 1-dimensional vector representation

control step as shown in Fig. 3. By solving the system differential equation in (6) over the $(1-m)T$ period and the mT period and eliminating the intermediate current vector $\mathbf{i}_{qd-m}^e[k]$, one obtains a discrete-time domain model with the delayed input (11).

$$\begin{aligned} \mathbf{i}_{qd}^e[k+1] &= \mathbf{A} \cdot \mathbf{i}_{qd}^e[k] \\ &+ \mathbf{B} \left\{ \boldsymbol{\mu}_i \cdot \mathbf{v}_{qd-m}^e[k] + (\mathbf{I} - \boldsymbol{\mu}_i) \mathbf{v}_{qd-m}^e[k] + \mathbf{v}_{qd}^e[k] \right\} \\ &- \mathbf{B}_e \cdot \mathbf{e}_{qd}^e[k] - \mathbf{B} \cdot \mathbf{d}_{qd-s}^e[k] - \mathbf{B}_e \cdot \mathbf{d}_{qd-e}^e[k], \end{aligned} \quad (11)$$

where $\boldsymbol{\mu}_i = \mathbf{B}^{-1} \mathbf{B}_m$ and

$$\mathbf{B}_m = \mathbf{L}^{-1} \cdot e^{-(r\mathbf{L}^{-1} + j\omega)mT} \int_{\tau=(1-m)T}^T e^{(r\mathbf{L}^{-1} + j\omega)\tau} \tau \cdot e^{-j\omega\tau} d\tau.$$

The discrete-time domain model has a moving-average input format, because the current step's input $\mathbf{v}_{qd-m}^e[k]$ and the past input $\mathbf{v}_{qd-m}^e[k-1]$ are averaged by the weighting matrix $\boldsymbol{\mu}_i$. The z-domain transfer function can also be written as

$$\begin{aligned} \mathbf{i}_{qd}^e(z) &= [\mathbf{I} - \mathbf{A}z^{-1}]^{-1} \left\{ \mathbf{B} [\boldsymbol{\mu}_i z^{-1} + (\mathbf{I} - \boldsymbol{\mu}_i) z^{-2}] \mathbf{v}_{qd-m}^e(z) \right. \\ &\left. + \mathbf{B} \cdot \mathbf{v}_{qd}^e(z) - \mathbf{B}_e \cdot \mathbf{e}_{qd}^e(z) - \mathbf{B} \cdot \mathbf{d}_{qd-s}^e(z) - \mathbf{d}_{qd-e}^e(z) \right\}. \end{aligned} \quad (12)$$

In the moving-average input format of the model, the voltage input from the controller cannot directly cancel the state feedback matrix $\mathbf{A}$. To obtain a model that enables plant pole cancellation, the system equation can be transformed into a moving-average output format. For symmetric AC machines, with the appropriate complex vector SISO model, the moving-average input block can simply be moved to the output. For asymmetric machines, transformation of the system model can be developed as follows. Assuming that the active state feedback through immediate input $\mathbf{v}_{qd}^e(z)$ is finished and the all the system matrices are modified accordingly and the back-EMF input is decoupled, the moving-average input format model can be transformed into a moving-average output format z transform model (13)

$$\mathbf{i}_{qd}^e(z) = [\boldsymbol{\mu} z^{-1} + (\mathbf{I} - \boldsymbol{\mu}) z^{-2}] [\mathbf{I} - \mathbf{A}_T z^{-1}]^{-1} \mathbf{B}_d \mathbf{v}_{qd-d}^e(z),$$

where $\mathbf{A}_T = \mathbf{B}_d \mathbf{B}^{-1} \mathbf{A} \mathbf{B} \mathbf{B}_d^{-1}$, $\mathbf{B}_d = \mathbf{B}_m + \mathbf{A}_m \mathbf{B} - \mathbf{B}_m \mathbf{B}^{-1} \mathbf{A} \mathbf{B}$, (13)

$$\mathbf{A}_m = \mathbf{L}^{-1} \cdot e^{-(r\mathbf{L}^{-1} + j\omega)mT} \mathbf{L}, \text{ and } \boldsymbol{\mu} = \mathbf{B}_m \mathbf{B}_d^{-1}$$

E. Discrete-Time Domain Model of a Symmetric Machine

In this section, the general AC machine model is used to develop the discrete-time domain model of a symmetric AC machine including the control delay. The vector variables and matrix coefficients in that model can be regarded as complex vectors and simple SISO methods can be utilized. Therefore the moving-average input format model (11) is sufficient and the complex coefficients (14) are derived.

$$\begin{aligned} \mathbf{A} &= \mathbf{A}_T = e^{-(r/L + j\omega)T}, \quad \mathbf{B}_e = \frac{1 - e^{-(r/L + j\omega)T}}{r + j\omega \mathbf{L}} \\ \mathbf{B} &= \mathbf{B}_d = \frac{1}{L} \cdot e^{-(r/L + j\omega)T} \int_{\tau=0}^T e^{(r/L)\tau} \tau \cdot e^{-j\omega\tau} d\tau = \frac{e^{-j\omega T} - e^{-(r/L + j\omega)T}}{r} \\ \boldsymbol{\mu}_i &= \boldsymbol{\mu} = \frac{e^{-j\omega mT} - e^{-(r/L + j\omega)mT}}{e^{-j\omega T} - e^{-(r/L + j\omega)T}} = e^{j\omega(1-m)T} \cdot \frac{1 - e^{-(r/L)mT}}{1 - e^{-(r/L)T}} \end{aligned} \quad (14)$$

The discrete-time domain transfer function of a symmetric machine with a control delay (15) can be written.

$$G_p(z) = \frac{e^{-j\omega T} - e^{-(r/L + j\omega)T}}{r} \cdot \frac{\mu z^{-1} + (1 - \mu) z^{-2}}{1 - e^{-(r/L + j\omega)T} z^{-1}} \quad (15)$$

III. DISCRETE-TIME DOMAIN CURRENT REGULATOR DESIGN FOR AC MACHINES

A. SISO Pole-Zero Cancellation (Symmetric case)

Starting from the discrete-time domain model of a symmetric machine (14), the complex vector current controller is designed using a SISO complex pole-zero cancellation approach. In this case, the complex vector system pole $e^{-(r/L + j\omega)T}$ is cancelled by placing a zero at the same location. However, the moving-average input zero at $-(1 - \mu)/\mu$, which is caused by the input delay and has a negative real value, will also show up in the closed loop transfer function, unless the controller cancels the zero by putting a pole at the same location. However, a z-domain pole with negative real value will cause forced oscillation, which is not generally acceptable. So, it is better to have the same moving-average input zero in the closed-loop response. Therefore, the desired closed loop response $G_d(z)$ will be set up as a first-order transfer function with the same moving-averaging input zero as shown in (16)

$$G_d(z) = \frac{1 - A_{cl}}{1 - A_{cl} z^{-1}} [\hat{\mu} z^{-1} + (\mathbf{I} - \hat{\mu}) z^{-2}] \text{ with } A_{cl} := e^{-2\pi f_{bw} T}, \quad (16)$$

where the closed loop pole A_{cl} is located to achieve the desired bandwidth f_{bw} . If the machine $G_p(z)$ itself is stable and if the moving-average input zeros are not cancelled in the desired closed loop response $G_d(z)$, a direct synthesis method for the discrete time controller $G_c(z)$ can be followed using (17).

$$G_c(z) = \frac{G_d(z)}{\hat{G}_p(z)(1 - G_d(z))} \quad (17)$$

Thus, the controller can be formulated as follows.

$$G_c(z) = \frac{r e^{j\omega T}}{1 - e^{-(\hat{r}/L)T}} \cdot \frac{(1 - e^{-(\hat{r}/L + j\omega)T} z^{-1}) (1 - A_{cl})}{1 - [A_{cl} + (1 - A_{cl}) \hat{\mu}] z^{-1} - (1 - A_{cl})(1 - \hat{\mu}) z^{-2}} \quad (18)$$

The rightmost term of (17) decouples the frame velocity dependent input coefficient $\mathbf{B}$. The controller also has a complex zero at $e^{-(\hat{r}/L + j\omega)T}$ that cancels the frame velocity dependent complex asymmetric machine pole. Similar to the integrator in the continuous-time domain PI regulator, the discrete-time domain controller has an accumulator using two step output memories. In case there is no control delay, thus $\mu = 0$, the accumulator becomes a pure accumulator with one-step output memory. The gain $(1 - A_{cl})$ is a normalized proportional gain that determines the discrete-time domain bandwidth. If $(1 - A_{cl}) = 1$, the controller will achieve dead-beat operation.

B. Multivariable Direct Control Synthesis (Asymmetric case)

The SISO pole-zero cancellation approach using a direct synthesis method can be generalized to asymmetric machines. Given an estimated discrete time model $\hat{G}_p(z)$ based on (11)

and the desired closed loop response $\mathbf{G}_d(z)$, the controller can be directly synthesized using (19) which is an extension of (17).

$$\mathbf{G}_c(z) = \hat{\mathbf{G}}_p(z)^{-1} \mathbf{G}_d(z) [\mathbf{I} - \mathbf{G}_d(z)]^{-1} \quad (19)$$

For a controller to be realizable, it can not require future input, and the controller itself should be stable. Even though the machine model $\mathbf{G}_p(z)$ is stable, due to the moving-average output block, the controller itself will be unstable or show forced oscillation, unless the $\mathbf{G}_d(z)$ is chosen so that the $\hat{\mathbf{G}}_p(z)^{-1} \mathbf{G}_d(z)$ neither has unstable poles nor poles with negative real values. Again, by choosing the desired response to have the same moving-average output block as the machine model including the delay, controller stability is achieved and forced-oscillation is avoided. For example, the desired closed-loop response can be set to be first order with the moving-average output block retained as in (20).

$$\mathbf{G}_d(z) = [\hat{\mu}z^{-1} + (\mathbf{I} - \hat{\mu})z^{-2}] (\mathbf{I} - \mathbf{A}_{cl}z^{-1})^{-1} (\mathbf{I} - \mathbf{A}_{cl}) \quad (20)$$

The closed-loop feedback matrix can be set following (21)

$$\mathbf{A}_{cl} = \text{diag} \left\{ e^{-2\pi f_{bw-q}T}, e^{-2\pi f_{bw-d}T} \right\}, \quad (21)$$

where f_{bw-q} and f_{bw-d} are the current regulation bandwidths for the q and d axes. Using the desired response and the plant model (22)

$$\hat{\mathbf{G}}_p(z) = [\hat{\mu}z^{-1} + (\mathbf{I} - \hat{\mu})z^{-2}] (\mathbf{I} - \hat{\mathbf{A}}_T z^{-1})^{-1} \hat{\mathbf{B}}_d, \quad (22)$$

the direct design controller can be synthesized as follows.

$$\mathbf{G}_c(z) = \hat{\mathbf{B}}_d^{-1} (\mathbf{I} - \hat{\mathbf{A}}_T z^{-1}) \left\{ \mathbf{I} - [\mathbf{A}_{cl} + (\mathbf{I} - \mathbf{A}_{cl})\hat{\mu}]z^{-1} - (\mathbf{I} - \mathbf{A}_{cl})(\mathbf{I} - \hat{\mu})z^{-2} \right\}^{-1} (\mathbf{I} - \mathbf{A}_{cl}) \quad (23)$$

Similar to the symmetric machine case, the rightmost term $\hat{\mathbf{B}}_d^{-1}$ is the input decoupling, the second term $(\mathbf{I} - \hat{\mathbf{A}}_T z^{-1})$ is the plant pole cancellation block, and the third one is the a two-step memory accumulator. The leftmost term is also the normalized discrete-time proportional gain.

C. Dead-beat Tracking Command Feedforward

The proposed controller turns into a dead-beat controller, if the $\mathbf{A}_{cl}$ is set to zero. However, this forces the accumulation block to cancel the moving-average output block, thus making the system have forced oscillatory dynamics. It is more realistic to tune the current feedback to an achievable bandwidth with sufficient stability margin and apply feedforward that achieves dead-beat command tracking. Assuming correct parameter estimates, to solve for the dead-beat command feedforward, the transfer function after the normalized proportional gain on the current is (24)

$$\mathbf{i}_{qd}^e(z) = [\hat{\mu}z^{-1} + (\mathbf{I} - \hat{\mu})z^{-2}] (\mathbf{I} - \mathbf{A}_{cl}z^{-1})^{-1} \tilde{\mathbf{v}}_{qd-cff}^e(z), \quad (24)$$

and by adding voltage command feedforward per (25)

$$\tilde{\mathbf{v}}_{qd-cff}^e(z) = (\mathbf{I} - z^{-1}) \mathbf{A}_{cl} \mathbf{i}_{qd}^{e*}(z) \quad (25)$$

the command tracking response achieved is dead-beat (26).

$$\mathbf{i}_{qd}^e(z) = [\hat{\mu}z^{-1} + (\mathbf{I} - \hat{\mu})z^{-2}] \mathbf{i}_{qd}^{e*}(z). \quad (26)$$

D. Immediate Virtual Resistance State Feedback

If we close the continuous-time domain model of a symmetric AC machine with a complex vector current regulator, the dynamic stiffness transfer function, the impedance of the motor winding to a disturbance voltage input, can be written as (27).

$$\frac{\mathbf{d}_{qd}^e(s)}{\mathbf{i}_{qd}^e(s)} = - \left[Ls + (r + K_p + j\omega) + K_p(\hat{r}/L + j\omega)s^{-1} \right] \quad (27)$$

Evaluating this function at the negative sequence of the frame rotation frequency $s = -j\omega$, which is the stationary reference frame dc disturbance seen from the synchronous reference frame, only the physical resistance and the integrated current state feedback term are left as shown in (28)

$$\frac{\mathbf{d}_{qd}^e(-j\omega)}{\mathbf{i}_{qd}^e(-j\omega)} = -r + \frac{K_i}{j\omega} \quad (28)$$

Therefore, as the fundamental frequency gets faster, the dynamic stiffness, the disturbance rejection impedance, around the stationary frame dc component gets close to just the physical resistance. Therefore, virtual resistance state feedback is needed to improve rejection of the stationary frame dc or slow frequency disturbances, which are not easy to decouple.

In case of digitally controlled AC machines, if virtual resistance state feedback is made on the immediate input (29)

$$\mathbf{v}_{qd}^e(z) = \hat{\mathbf{B}}^{-1} [\mathbf{A}_v - \hat{\mathbf{A}}] \mathbf{i}_{qd}^e(z), \quad (29)$$

where $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$ are the estimated system feedback matrix and input matrix respectively, and $\mathbf{A}_v$ is the desired system feedback matrix. The modified plant in moving-average output format becomes (30)

$$\mathbf{i}_{qd}^e(z) = [\hat{\mu}z^{-1} + (\mathbf{I} - \hat{\mu}_v)z^{-2}] [\mathbf{I} - \mathbf{A}_v z^{-1}]^{-1} \mathbf{B}_v \mathbf{v}_{qd-d}^e(z),$$

where $\mathbf{A}_v T = \mathbf{B}_v T^{-1} \mathbf{A} T \mathbf{B}_v^{-1}$, $T = \mathbf{A}_v \mathbf{B}_m + \mathbf{A}_m \mathbf{B}_{1-m}$, (30)

$$\mathbf{B}_v = \mathbf{B}_m + \mathbf{A}_v^{-1} \mathbf{A}_m \mathbf{B}_{1-m} T^{-1} \mathbf{A}_v T, \text{ and } \hat{\mu}_v = \mathbf{B}_m \mathbf{B}_v^{-1}$$

This model can directly be used with the controller synthesis formula in (23).

E. Delayed Virtual Resistance State Feedback

If the controller cannot implement immediate input to the machine, virtual resistance state feedback will be implemented by adding it to the delayed control input $\mathbf{v}_{qd-m-c}[k]$ as in (31)

$$\mathbf{v}_{qd-m}^e(z) = \mathbf{v}_{qd-m-c}^e(z) + \hat{\mathbf{B}}_d^{-1} \mathbf{A}_{dv} \mathbf{i}_{qd}^e(z), \quad (31)$$

and the modified plant model would become (32)

$$\mathbf{i}_{qd}^e(z) = [\hat{\mu}z^{-1} + (\mathbf{I} - \hat{\mu})z^{-2}] \left\{ \mathbf{I} - (\mathbf{A}_T + \hat{\mu} \mathbf{A}_{dv})z^{-1} - (\mathbf{I} - \hat{\mu}) \mathbf{A}_{dv} z^{-2} \right\}^{-1} \mathbf{B}_d \mathbf{v}_{qd-d}^e(z). \quad (32)$$

Since the state feedback of the model has another delay, the controller has to be reformulated. Solving for the controller that meets the same desired response transfer function in (20), the controller can be solved as follows.

$$\mathbf{G}_c(z) = \hat{\mathbf{B}}_d^{-1} [\mathbf{I} - (\hat{\mathbf{A}} + \hat{\mu} \mathbf{A}_{dv})z^{-1} - (\mathbf{I} - \hat{\mu}) \mathbf{A}_{dv} z^{-2}] \left\{ \mathbf{I} - [\mathbf{A}_{cl} + (\mathbf{I} - \mathbf{A}_{cl})\hat{\mu}]z^{-1} - (\mathbf{I} - \mathbf{A}_{cl})(\mathbf{I} - \hat{\mu})z^{-2} \right\}^{-1} (\mathbf{I} - \mathbf{A}_{cl}) \quad (33)$$

The generic control block diagram is drawn in Fig. 4.

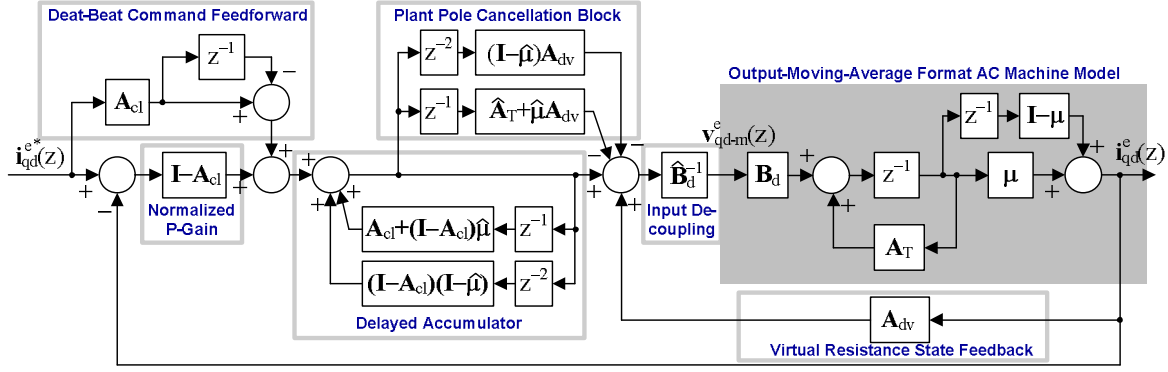


Fig. 4. Proposed discrete-time domain current regulation system for AC machines with delayed virtual resistance state feedback

IV. SIMULATIONAL EVALUATION

A. Comparison to Digital Implementations of Continuous-Time Domain Designs

Continuous-time domain design of the synchronous frame PI and the complex vector current regulator will be compared to the discrete-time domain design proposed. Using the backward difference method, the synchronous frame PI current (SFPI) regulator was implemented as a discrete-time domain controller following (34)

$$v_{qd}^e(z) = K_p \left(1 + \frac{(K_i/K_p)T}{1-z^{-1}} \right) (i_{qd}^*(z) - i_{qd}^e(z)) + j\omega \hat{L} i_{qd}^e(z), \quad (34)$$

and the complex vector current regulator (CVCR) was discretized and implemented following (35)

$$v_{qd}^e(z) = K_p \left(1 + \frac{(K_i/K_p + j\omega)T}{1-z^{-1}} \right) (i_{qd}^*(z) - i_{qd}^e(z)). \quad (35)$$

The compensation method of [7] for the magnitude attenuation and the phase lag of the controller output voltage caused by the control delay and the PWM-VSI was included in the simulation model.

The discrete-time complex vector root loci of the proposed controller and the two digital implementations above are

calculated and shown in Fig. 5. For simplicity, the normalized proportional gain is defined as $K = (1 - A_{cl})$. For the digital implementations in (34) and (35), K corresponds to $(K_p + K_i T)/B$. For the root loci plot, the normalized p-gain K is varied within the range 0 to 1, where 1 corresponds to dead-beat operation. In order to provide general results independent of the sampling frequency, f_s , the discrete-time domain poles and zeros and the electrical rotational frequency f_1 are presented relative to f_s .

In the computation, controller delay is set as one-step and electrical rotational frequency f_1 is set at a fairly high speed $0.05 f_s$ (500Hz for 10kHz sampling) and the estimated inductance used for design is 25% overestimated (more than the real inductance) to show robustness of the designs. The electric time constant is chosen to be 24 times the sampling period T (2.4ms for 10kHz sampling).

The synchronous frame PI (SFPI) current regulator with cross coupling decoupling shows poor performance with big oscillations, and is unstable if K is bigger than 0.58. Secondly, the complex vector current regulator shows better robustness and performance and maintains good non-oscillatory response until K is 0.20 reaching the bandwidth of $0.1 f_s$ Hz. However, it also becomes unstable if K is bigger than 0.8, which is about 40% more robustness margin than the synchronous frame PI. In case of the proposed method, the controller stayed stable up

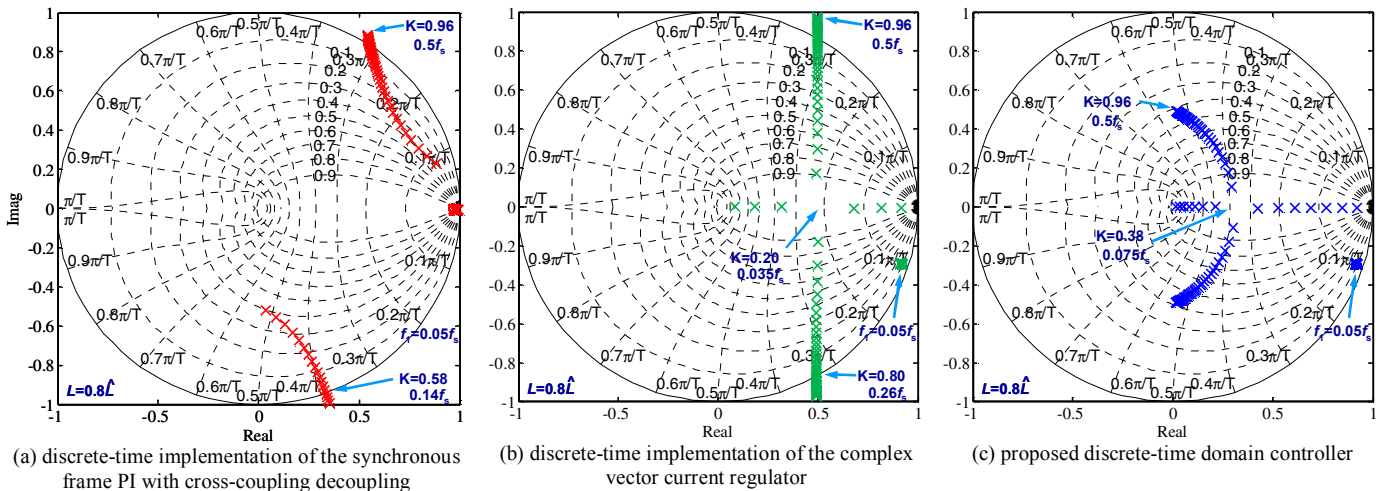


Fig. 5. Comparison of discrete-time complex vector root locus of the current regulators implemented in a 10kHz control loop with 1-step control delay at 500Hz fundamental frequency operation with $L=0.8\hat{L}$

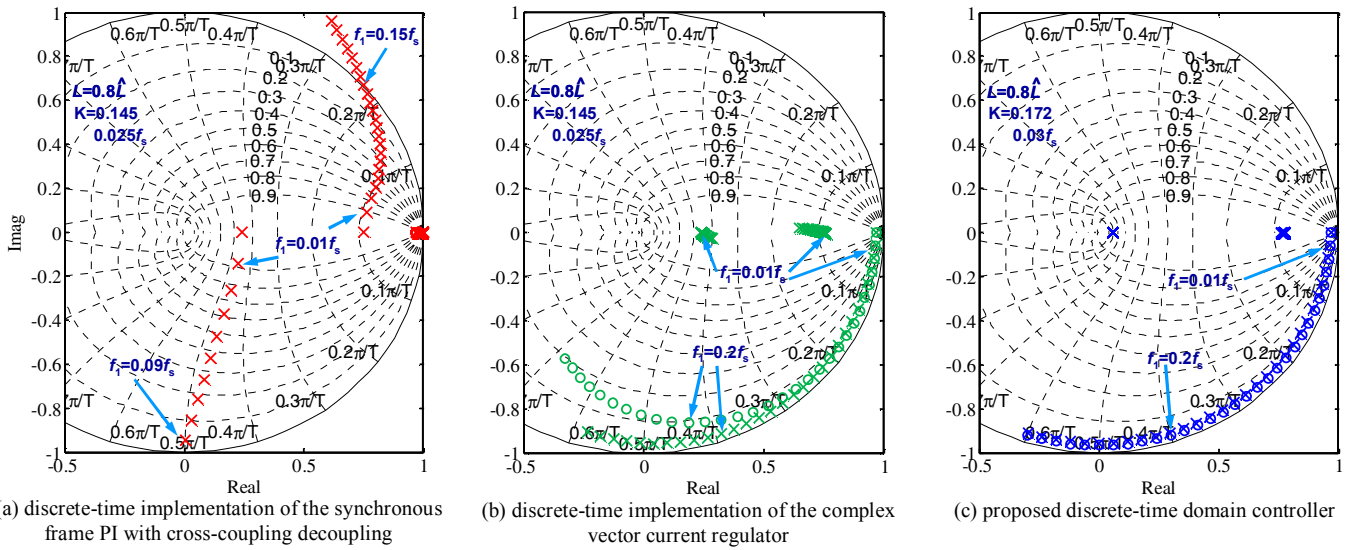


Fig. 6. Comparison of discrete-time complex vector root migration plot with respect to fundamental frequencies at 0,100,...,3000Hz of the current regulators implemented in a 10kHz control loop with 1-step control delay and tuned at 300Hz bandwidth for $L=0.8\hat{L}$

to the deadbeat operation $K=1$ with enough robustness margin and the system maintains non-oscillatory response up to $0.2f_s$ Hz where K is 0.38. Even with bigger K , the oscillation didn't exceed the damping ratio 0.4.

B. Compariosn to Digital Implementations of Continuous-Time Domain Designs : High Speed Performance

To compare performance at high fundamental frequencies, a complex vector root migration plot with respect to the fundamental frequency variation is computed up to 30% of the sampling frequency and provided in Fig. 6. All three controllers are tuned at 300Hz bandwidth. The estimated inductance again is 25% overestimated. The SFPI gets oscillatory as the machine speed increases and become unstable if the fundamental frequency is bigger than $0.09f_s$ Hz. The CVCR remains stable up to $0.3f_s$ Hz. However, as the frequency gets higher the complex pole cancellation zero deviates from the pole location, which causes high frequency oscillation. By comparison, the proposed controller puts the cancellation complex zero very close to the machine's complex

pole, which makes the current tracking response very robust.

C. Effect of Virtual Resistance State Feedback and Improvement with Immediate VRSF

The effect of virtual resistance state feedback on the closed loop performance of the proposed discrete-time domain controller is investigated via complex vector root migration plots. The proposed current regulator was tuned at $0.1f_s$ Hz and the machine was run at $0.1f_s$ Hz. The virtual resistance state feedback (VRSF) was tuned within the range $f_v=0\sim 0.08f_s$ and varied, where $f_v=2\pi r_v L$ and r_v is the virtual resistance. With ideal estimates, the command responses are not affected by the VRSF tuned as shown in Fig. 7 (a). However, with inaccurate estimates of L , the delayed VRSF makes the command tracking sluggish and oscillatory, thus limiting the achievable bandwidth as seen in Fig. 7 (b). By comparison, the immediate VRSF affected the command tracking only slightly as shown in Fig. 7 (c). Therefore, immediate VRSF appears to be the most desirable implementation.

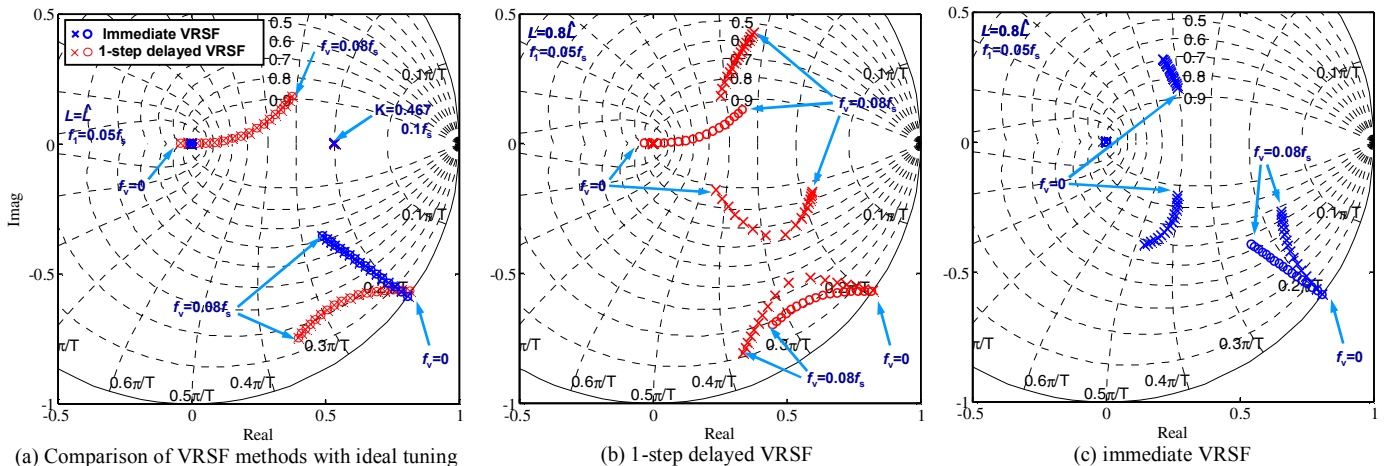
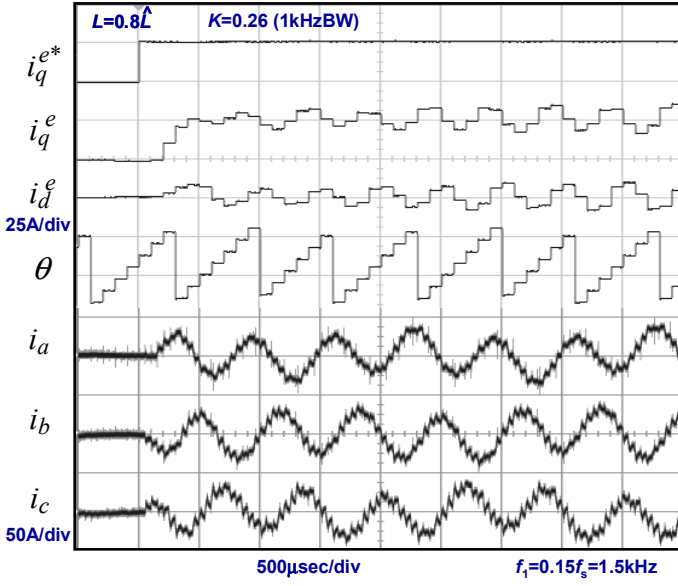
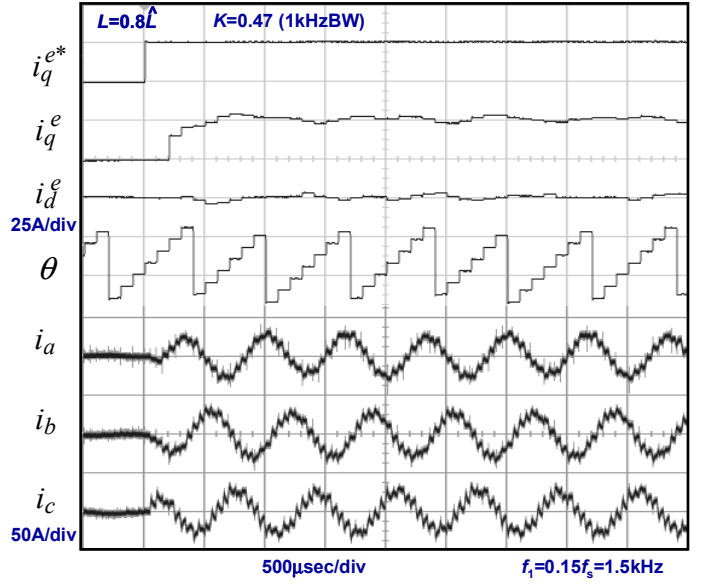


Fig. 7. Comparison of discrete-time complex vector root migration plot with respect to the variation of the VRSF pole tuned at 0.50,...,800Hz of the proposed discrete-time domain current regulator implemented in a 10kHz control loop with 1-step control delay and tuned at 1000Hz bandwidth



(a) discrete-time implementation of the complex vector current regulator



(b) proposed discrete-time domain controller

Fig. 8. Comparison of step current command response of the 1-step delayed current regulators tuned at 1kHz with inaccurate parameter $L=0.8\hat{L}$

V. EXPERIMENTAL EVALUATION

An AC machine controller using the TI TMS320C6713 DSP and Xilinx Virtex-E FPGA was used for the experimental evaluation. The controller drives a PMSM through a Semikron IPM module with 10kHz SVPWM. The controller can implement both one-step and half-step delay operation. For evaluation of high speed operation, a 3 phase R-L load that matches the motor parameters was used. The machine parameters used in the experiment are shown in Table 1.

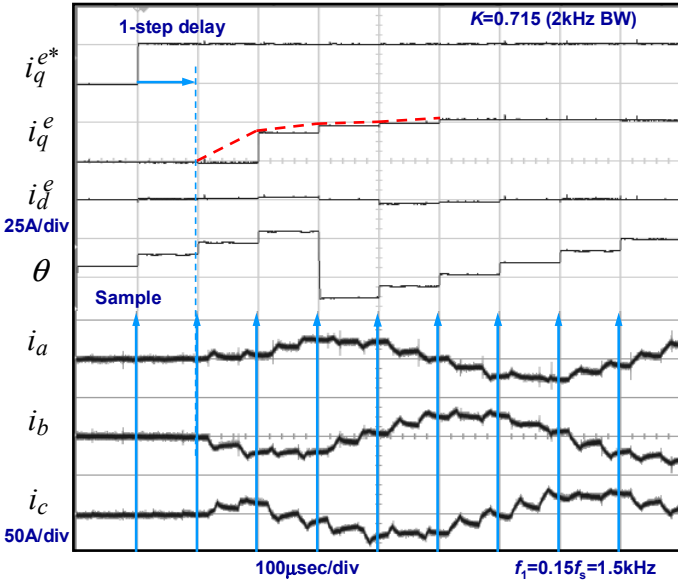
A. Robustness Comparison

Step current command response of the proposed discrete-time domain current regulator is compared to the discrete-time implementation of the CVCR as shown in Fig. 8. The control

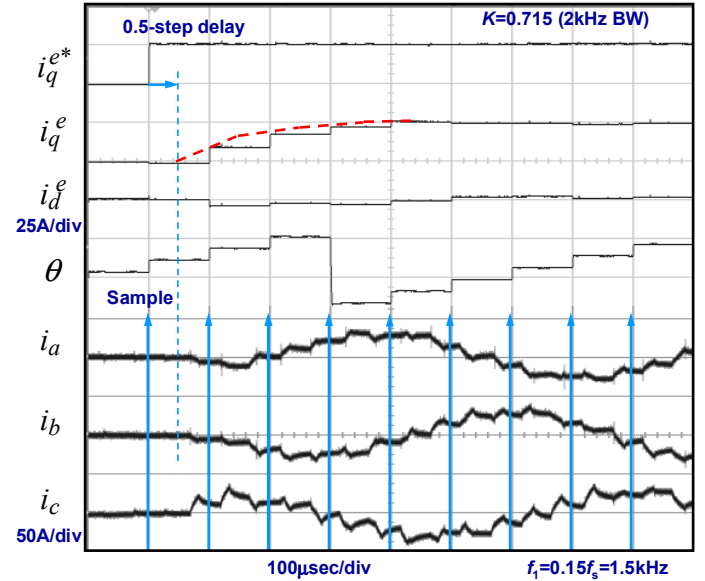
loop is also implemented with 10kHz sampling with 1-step control delay and the current regulators are tuned at 1kHz with 300Hz VRSF with 25% overestimated inductance. At fundamental frequency of 1.5kHz, a step command of 25A is applied to the q-axis. The discrete-time domain CVCR show significant oscillation, whereas the proposed controller shows fast, but well-damped transient properties.

B. Transient Response of the Proposed Controller

Transient responses of the 1-step delay and half-step delay implementations of the proposed current regulators tuned at 2kHz bandwidth with 300Hz VRSF are tested. As shown in Fig. 9, in both cases, the proposed current regulator, operating at 1.5kHz fundamental frequency, tracks the step current command at the desired bandwidth with the inherent control



(a) proposed discrete-time domain controller with a 1-step delay



(b) proposed discrete-time domain controller with a half-step delay

Fig. 9. Transient response of the proposed current regulator with 1-step delay and 0.5-step delay implementation tuned at 2kHz bandwidth

delay. In case of the half-step delayed system, the phase currents are sampled in the middle of one PWM-cycle, thus showing small first step magnitude. The red dashed lines are the approximate response envelopes of the actual current.

To the same 1-step delayed controller, at the same operating conditions, dead-beat command feedforward derived in (25) is applied and the current tracking response is shown in Fig. 10. With the proposed discrete-time domain current regulator based on the accurate modeling along with the exact system parameters, the dead-beat command feedforward operated very nicely and the current tracked the command in one-step after the voltage is applied. The dead-beat command feedforward was implemented using only a simple gain and a delay unit. This proves that the proposed discrete-time domain modeling method is accurate and the proposed discrete-time domain current regulator is designed to handle the high speed operation very well.

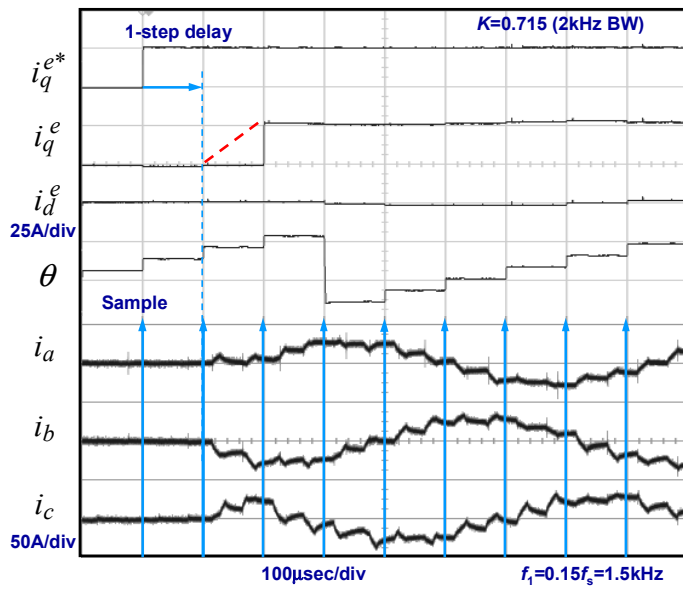


Fig. 10. Dead-beat command tracking response of the proposed discrete-time current regulator with 1-step delay implementation tuned at 2kHz bandwidth

TABLE 1
MACHINE AND PWM-VSI DATA

Parameter	Value
Rated output power	6.3 kW
Rated current	62A _{rms}
Rated torque / speed	6.7 Nm @ 9000rpm
Maximum Speed	15,000 rpm
Number of poles	4
Number of phases	3
Phase resistances	0.040 Ω
Q-axis inductance	95 µH
D-axis inductance	65 µH
PM Flux Linkage	0.0378 Wb
Switching Frequency	10 kHz

VI. CONCLUSION

A discrete-time domain model of AC machines, which includes the stationary frame voltage latch effect of a PWM-VSI, the synchronous frame voltage latch effect of the back-EMF, and the digital control delay effect, including fractional-step delay, is developed for general AC machines, including both symmetric and asymmetric machines.

Based on the proposed models, discrete-time domain current regulation methods are proposed. The primary design proposed is a generalization of the complex vector pole-zero cancellation approach. The proposed current regulator explicitly provides a tool to tune the command tracking bandwidth. The secondary design achieves dead-beat command tracking via a command feedforward solution.

The discrete-time domain model was shown to enable discrete-time domain complex vector analyses, such as root locus and root migration analysis. Comparative analysis suggested limitations in stability and tracking performance of the conventional implementations of synchronous frame PI and complex vector methods in terms of current tracking bandwidth and operation speed, and proved superior capability of the proposed discrete-time domain current regulator which correctly compensated the fractional step delay.

Experimental evaluation supported the analysis and proved robustness of the proposed current regulator tuned at high current regulation bandwidth and operating under high speed conditions, and showed good dead-beat command tracking response with the addition of a simple command feedforward to the proposed discrete-time domain current regulator.

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